Within the flat 6D Conformal Spacetime Algebra system ($Cl(4,1,1)$), we can construct a complex vector space (Hilbert space) to represent local realist probability theory.

In this representation, probabilities are weighted by complex amplitudes, where the modulus squared of the amplitude corresponds to the classical probability of a state, and the phase corresponds to the physical precessional phase of the local soliton current loops [13.2].

Section 14: Construction of the Complex Hilbert Space and Quadratic Forms

This section mathematically formalizes the probability state space of the system as a complex vector space and extends it into a tensor space of row and column matrices to construct physical quadratic forms.

14.1 Complex Vector Space of Probability Amplitudes We construct a complex vector space $\mathcal{H} \cong \mathbb{C}^N$ to represent the probability distribution of the N mutually exclusive physical configurations of our system (such as the four states $[00], [01], [10], [11]$ of the Josephson-junction register [Section 13]).

Any state of the system is represented as a linear combination of the basis elements $\{|k\rangle\}$:

$$|\Psi\rangle = \sum_{k=1}^{N} c_k |k\rangle$$

where $c_k \in \mathbb{C}$ are the complex probability weights. Each complex weight can be written in polar form:

$$c_k = \sqrt{p_k} e^{i\varphi_k}$$

where: * $p_k = |c_k|^2 \in [0,1]$ represents the classical probability of the system being physically found in configuration k. * φ_k represents the physical precessional phase of the local soliton current loops [13.2].

The conservation of total probability requires the normalization condition:

$$\sum_{k=1}^{N} |c_k|^2 = 1$$

14.2 Tensor Space Representation: Column and Row Matrices We represent the vector state $|\Psi\rangle$ as a column matrix $\mathbf{\Psi}$ in the tensor space:

$$\mathbf{\Psi} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}$$

The dual space of this vector space consists of row matrices. The dual vector $\langle\Psi|$ is represented by the conjugate transpose (adjoint) of the column matrix $\mathbf{\Psi}$:

$$\mathbf{\Psi}^\dagger = \begin{pmatrix} c_1^* & c_2^* & \cdots & c_N^* \end{pmatrix}$$

The inner product of two states $|\Phi\rangle = \sum a_k|k\rangle$ and $|\Psi\rangle = \sum c_k|k\rangle$ is the matrix product of the dual row and the column matrices:

$$\langle\Phi|\Psi\rangle = \mathbf{\Phi}^\dagger\mathbf{\Psi} = \begin{pmatrix} a_1^* & a_2^* & \cdots & a_N^* \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix} = \sum_{k=1}^{N} a_k^* c_k$$

14.3 Quadratic Forms via Matrix Sandwiching Let $\mathbf{M}$ be an $M \times N$ rectangular matrix representing a physical transition or coupling property of the measuring system. We construct a quadratic (or bilinear) form by sandwiching the matrix $\mathbf{M}$ between a dual row vector $\mathbf{\Phi}^\dagger$ of dimension $1 \times M$ and a column vector $\mathbf{\Psi}$ of dimension $N \times 1$:

$$\mathcal{Q}(\mathbf{\Phi}, \mathbf{\Psi}) = \mathbf{\Phi}^\dagger\mathbf{M}\mathbf{\Psi}$$

In the case of a self-measurement, $\mathbf{\Phi} = \mathbf{\Psi}$, and $\mathbf{M}$ is a square $N \times N$ matrix. The quadratic form yields a real scalar representing the expected value of the physical measurement (such as the joint SQUID current correlation ρ [13.4]):

$$\rho = \mathbf{\Psi}^\dagger\mathbf{M}\mathbf{\Psi}$$

Expanding this matrix sandwich product yields:

$$\rho = \begin{pmatrix} c_1^* & c_2^* & \cdots & c_N^* \end{pmatrix} \begin{pmatrix} M_{11} & M_{12} & \cdots & M_{1N} \\ M_{21} & M_{22} & \cdots & M_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ M_{N1} & M_{N2} & \cdots & M_{NN} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}$$

$$\rho = \sum_{i=1}^{N} \sum_{j=1}^{N} c_i^* M_{ij} c_j$$

This sandwich structure $\mathbf{\Psi}^\dagger \mathbf{M} \mathbf{\Psi}$ mathematically represents the complete algebraic projection of the system's precessional state onto the physical measuring device's coupling matrix $\mathbf{M}$. This recovers the identical expectation values of quantum probability using entirely classical wave mechanics and local precessional phases.